Fractions, Decimals and Percentages

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · convert *(panoni)*

A Percentages and decimals ●○○

⚠ **Watch out:** Per cent means "out of 100", so divide by 100: 8% is 0.08, not 0.8.

1. Write each percentage as a decimal.

a) 50% = _____ **b)** 25% = _____ **c)** 8% = _____ **d)** 70% = _____

e) 3% = _____ **f)** 120% = _____ **g)** 12.5% = _____ **h)** 0.5% = _____

2. Write each decimal as a percentage.

a) 0.45 = _____ **b)** 0.7 = _____ **c)** 0.09 = _____ **d)** 0.3 = _____

e) 1.5 = _____ **f)** 0.125 = _____ **g)** 0.01 = _____ **h)** 2 = _____

B Fractions to decimals and percentages ●●○

Remember: A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. Then multiply by 100 for the percentage: 75%.

3. Write each fraction as a decimal.

a) $\frac{1}{2}$ = _____ **b)** $\frac{3}{4}$ = _____ **c)** $\frac{2}{5}$ = _____ **d)** $\frac{7}{10}$ = _____

e) $\frac{1}{8}$ = _____ **f)** $\frac{9}{20}$ = _____ **g)** $\frac{3}{25}$ = _____ **h)** $\frac{3}{2}$ = _____

4. Write each fraction as a percentage.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{5}$ = _____ **c)** $\frac{3}{10}$ = _____ **d)** $\frac{4}{5}$ = _____

e) $\frac{11}{20}$ = _____ **f)** $\frac{9}{25}$ = _____ **g)** $\frac{7}{50}$ = _____ **h)** $\frac{5}{4}$ = _____

C Writing fractions ●●○

Remember: Write the percentage over 100, then simplify: $40\% = \frac{40}{100} = \frac{2}{5}$.

5. Write each percentage as a fraction in its simplest form.

a) 50% = _____ **b)** 25% = _____ **c)** 40% = _____ **d)** 90% = _____

e) 15% = _____ **f)** 64% = _____ **g)** 5% = _____ **h)** 1% = _____

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

6. Write each decimal as a fraction in its simplest form.

a) 0.3 = _____ **b)** 0.5 = _____ **c)** 0.25 = _____ **d)** 0.8 = _____

e) 0.45 = _____ **f)** 0.06 = _____ **g)** 0.125 = _____ **h)** 0.72 = _____

D All three together ●●○

7. Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
	0.25	
		10%
$\frac{3}{5}$		
	0.45	
		8%
$\frac{3}{8}$		

E Comparing and ordering ●●●

⚠ **Watch out:** Do not compare the digits you can see. Change everything to the same form first (percentages are usually the easiest).

8. Which is larger?

a) $\frac{1}{2}$ or 0.45 **b)** 30% or $\frac{1}{4}$ **c)** 0.7 or 68%

d) $\frac{3}{5}$ or 65% **e)** 0.08 or $\frac{1}{10}$ **f)** $\frac{3}{8}$ or 40%

9. Write each list in order, smallest first.

a) $\frac{1}{2}, \quad 0.4, \quad 45\%$ **b)** $70\%, \quad \frac{3}{4}, \quad 0.72$

c) $8\%, \quad 0.7, \quad \frac{3}{4}, \quad 72\%$ **d)** $\frac{9}{10}, \quad 0.95, \quad 89\%, \quad \frac{4}{5}$

F Problems ●●●

10. Aroha scores 18 out of 25 in a test. Ben scores 70% in the same test. Who did better? Show your working.

11. $\frac{3}{5}$ of the students in a class play a sport. What percentage of the class do not play a sport?

★ Extension

12. Write 0.5% as a fraction in its simplest form.

13. Write 250% as a mixed number in its simplest form.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Fractions, Decimals and Percentages

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · convert *(panoni)*

A Percentages and decimals ●○○

⚠ **Watch out:** Per cent means "out of 100", so divide by 100: 8% is 0.08, not 0.8.

1. Write each percentage as a decimal.

a) 75% = _____ **b)** 40% = _____ **c)** 6% = _____ **d)** 90% = _____

e) 2% = _____ **f)** 150% = _____ **g)** 37.5% = _____ **h)** 0.2% = _____

2. Write each decimal as a percentage.

a) 0.35 = _____ **b)** 0.9 = _____ **c)** 0.04 = _____ **d)** 0.6 = _____

e) 1.2 = _____ **f)** 0.375 = _____ **g)** 0.07 = _____ **h)** 3 = _____

B Fractions to decimals and percentages ●●○

💡 **Remember:** A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. Then multiply by 100 for the percentage: 75%.

3. Write each fraction as a decimal.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{5}$ = _____ **c)** $\frac{3}{5}$ = _____ **d)** $\frac{9}{10}$ = _____

e) $\frac{3}{8}$ = _____ **f)** $\frac{7}{20}$ = _____ **g)** $\frac{6}{25}$ = _____ **h)** $\frac{5}{2}$ = _____

4. Write each fraction as a percentage.

a) $\frac{1}{2}$ = _____ **b)** $\frac{3}{4}$ = _____ **c)** $\frac{7}{10}$ = _____ **d)** $\frac{2}{5}$ = _____

e) $\frac{13}{20}$ = _____ **f)** $\frac{8}{25}$ = _____ **g)** $\frac{9}{50}$ = _____ **h)** $\frac{7}{4}$ = _____

C Writing fractions ●●○

💡 **Remember:** Write the percentage over 100, then simplify: $40\% = \frac{40}{100} = \frac{2}{5}$.

5. Write each percentage as a fraction in its simplest form.

a) 75% = _____ **b)** 20% = _____ **c)** 60% = _____ **d)** 30% = _____

e) 35% = _____ **f)** 48% = _____ **g)** 4% = _____ **h)** 1% = _____

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

6. Write each decimal as a fraction in its simplest form.

a) 0.7 = _____ **b)** 0.2 = _____ **c)** 0.75 = _____ **d)** 0.4 = _____

e) 0.35 = _____ **f)** 0.08 = _____ **g)** 0.375 = _____ **h)** 0.64 = _____

D All three together ●●○

7. Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
	0.5	
		20%
$\frac{4}{5}$		
	0.35	
		12%
$\frac{5}{8}$		

E Comparing and ordering ●●●

> ⚠ **Watch out:** Do not compare the digits you can see. Change everything to the same form first (percentages are usually the easiest).

8. Which is larger?

a) $\frac{1}{2}$ or 0.55 **b)** 20% or $\frac{1}{4}$ **c)** 0.6 or 58%

d) $\frac{2}{5}$ or 45% **e)** 0.09 or $\frac{1}{10}$ **f)** $\frac{5}{8}$ or 60%

9. Write each list in order, smallest first.

a) $\frac{1}{2}, \quad 0.6, \quad 55\%$ **b)** $80\%, \quad \frac{3}{4}, \quad 0.78$

c) $6\%, \quad 0.5, \quad \frac{11}{20}, \quad 52\%$ **d)** $\frac{7}{10}, \quad 0.75, \quad 69\%, \quad \frac{4}{5}$

F Problems ●●●

10. Aroha scores 21 out of 25 in a test. Ben scores 80% in the same test. Who did better? Show your working.

11. $\frac{3}{4}$ of the students in a class play a sport. What percentage of the class do not play a sport?

★ Extension

12. Write 2.5% as a fraction in its simplest form.

13. Write 350% as a mixed number in its simplest form.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.